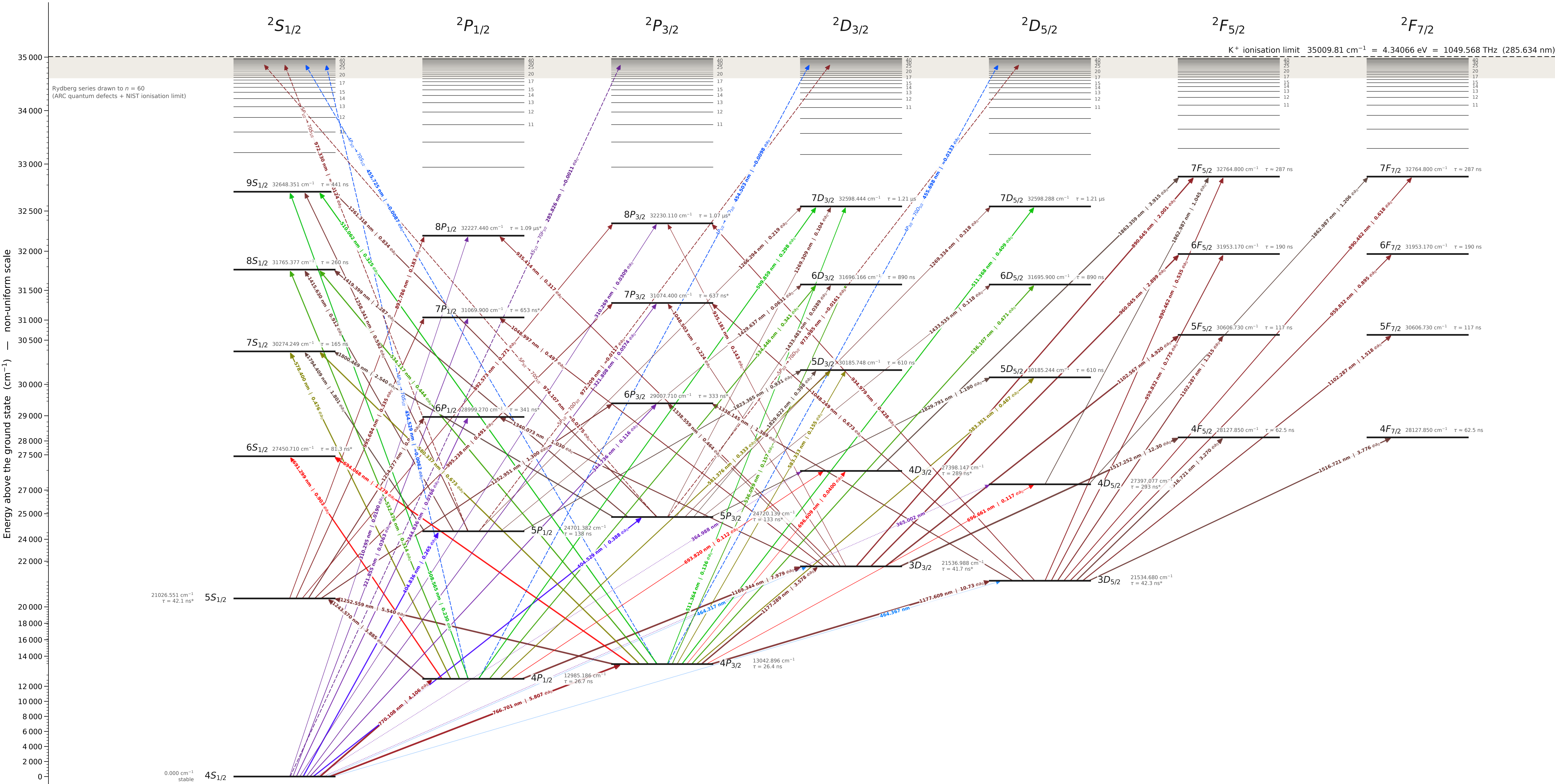
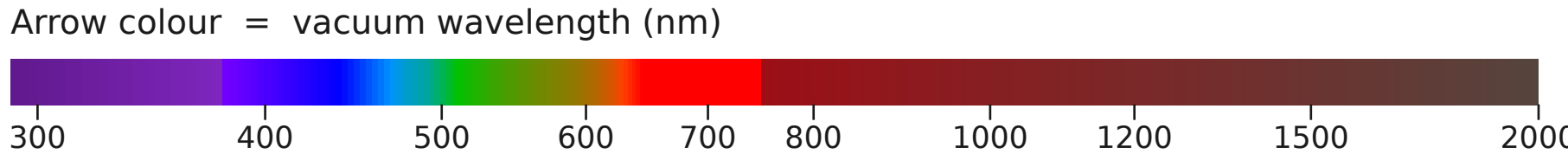


Potassium-39

³⁹K (K I), nuclear spin $I = 3/2$. Energy levels and transitions below 2 μm : 34 levels, 84 lines.

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle J||er||J' \rangle$ in ea0
Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: NIST energy and lifetime (measured where a value exists, otherwise ARC, marked $\approx$).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\tilde{\nu}$ (cm ⁻¹)	$\langle J er J' \rangle$ (ea0)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Typical use
$4S_{1/2} - 4P_{3/2}$	766.7009	766.4899	391.0162	13042.896	5.807	3.790e+07	6.031	100.0	D2 line
$4S_{1/2} - 4P_{1/2}$	770.1084	769.8965	389.2861	12985.185	4.106	3.740e+07	5.952	100.0	D1 line
$4P_{3/2} - 3D_{5/2}$	1177.6089	1177.2866	254.5773	8491.784	10.73	2.383e+07	3.764	100.0	
$4P_{1/2} - 3D_{3/2}$	1169.3442	1169.0241	256.3766	8551.802	7.979	2.017e+07	3.814	83.5	
$4P_{3/2} - 5S_{1/2}$	1252.5591	1252.2165	239.3440	7983.655	5.540	1.582e+07	3.780	66.5	
$4P_{1/2} - 5S_{1/2}$	1243.5699	1243.2298	241.0741	8041.365	3.885	7.951e+06	3.780	33.5	
$4P_{3/2} - 6S_{1/2}$	694.0678	693.8764	431.9354	14407.814	1.279	4.956e+06	1.959	40.3	
$4P_{3/2} - 3D_{3/2}$	1177.2889	1176.9667	254.6465	8494.092	3.578	3.974e+06	3.814	16.5	
$4P_{1/2} - 6S_{1/2}$	691.2988	691.1081	433.6655	14465.525	0.903	2.500e+06	1.959	20.3	
$4P_{3/2} - 7S_{1/2}$	580.3375	580.1766	516.5830	17231.353	0.673	2.348e+06	0.965	35.0	
$4P_{3/2} - 8S_{1/2}$	534.1173	533.9687	561.2858	18722.481	0.444	1.311e+06	0.612	32.9	
$4P_{1/2} - 7S_{1/2}$	578.4003	578.2400	518.3131	17289.063	0.476	1.186e+06	0.965	17.7	
$4S_{1/2} - 5P_{3/2}$	404.5285	404.4142	741.0911	24720.139	0.388	1.150e+06	1.195	16.8	404.5 nm blue line: narrow-line
$4S_{1/2} - 5P_{1/2}$	404.8356	404.7213	740.5288	24701.382	0.265	1.070e+06	1.157	15.8	404.8 nm blue line
$4P_{3/2} - 9S_{1/2}$	510.0621	509.9200	587.7568	19605.455	0.325	8.064e+05	0.361	31.8	
$4P_{1/2} - 8S_{1/2}$	532.4759	532.3279	563.0160	18780.191	0.314	6.620e+05	0.612	16.6	
$4P_{3/2} - 6D_{5/2}$	536.1067	535.9576	559.2030	18653.004	0.471	4.860e+05	0.179	45.1	

λ and ν from NIST level energies (centre of gravity, no hyperfine structure); λ air uses standard air (Ciddor).
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$. Branch: share of the upper level's decays that takes this line.

Rydberg excitation wavelengths (vacuum, nm)

Lower $\rightarrow$ series	$n = 30$	$n = 40$	$n = 50$	$n = 70$	$n = 100$	$n \rightarrow \infty$	d_{70} ARC
$4S_{1/2} \rightarrow nP_{3/2}$	286.757	286.246	286.019	285.826	285.727	285.634	0.0011
$4P_{3/2} \rightarrow nS_{1/2}$	458.187	456.825	456.227	455.725	455.468	455.230	0.0087
$4P_{3/2} \rightarrow nD_{5/2}$	457.819	456.676	456.152	455.698	455.459	455.230	0.0133
$5S_{1/2} \rightarrow nP_{3/2}$	722.223	718.989	717.556	716.346	715.722	715.141	0.0039
$3D_{5/2} \rightarrow nP_{3/2}$	749.737	746.253	744.708	743.406	742.734	742.108	0.0079
$3D_{3/2} \rightarrow nF_{7/2}$	748.888	745.906	744.534	743.343	742.713	742.108	0.0555
$5P_{3/2} \rightarrow nS_{1/2}$	985.427	979.149	976.402	974.107	972.932	971.848	0.0175
$5P_{3/2} \rightarrow nD_{5/2}$	983.722	978.461	976.058	973.985	972.891	971.848	0.0161
$4D_{3/2} \rightarrow nP_{3/2}$	1337.683	1326.632	1321.759	1317.661	1315.551	1313.588	0.0205
$4D_{5/2} \rightarrow nF_{7/2}$	1334.983	1325.536	1321.209	1317.465	1315.485	1313.588	0.0898
$6S_{1/2} \rightarrow nP_{3/2}$	1347.350	1336.139	1331.196	1327.039	1324.899	1322.908	0.0083
$4F_{7/2} \rightarrow nD_{5/2}$	1479.780	1467.907	1462.506	1457.855	1455.407	1453.074	0.0176
$6P_{3/2} \rightarrow nS_{1/2}$	1706.393	1687.655	1679.510	1672.731	1669.272	1666.082	0.0287
$6P_{3/2} \rightarrow nD_{5/2}$	1701.288	1685.613	1678.495	1672.372	1669.151	1666.082	0.0173

Rydberg term energies from ARC quantum defects + NIST ionisation limit, lower level from NIST. $d_{70} = \langle J||er||J' \rangle$ to $n = 70$ in ea0 (calculated).
 $S \rightarrow nP$ values come from a near-cancelling integral: treat them as an order of magnitude.

³⁹K hyperfine structure ($I = 3/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$4S_{1/2}$	230.859860	-	F=1: -288.57 F=2: +173.15
$4P_{1/2}$	27.79(7)	-	F=1: -34.74 F=2: +20.84
$4P_{3/2}$	6.084(17)	2.84(4)	F=0: -19.26 F=1: -16.02 F=2: -6.69 F=3: +14.40
$5S_{1/2}$	55.5(6)	-	F=1: -69.38 F=2: +41.62
$3D_{5/2}$	-0.62(4)	-	F=1: +3.25 F=2: +2.02 F=3: +0.15 F=4: -2.33
$3D_{3/2}$	0.96(4)	0.37(8)	F=0: -3.14 F=1: -2.55 F=2: -1.00 F=3: +2.25
$5P_{1/2}$	9.01(17)	-	F=1: -11.26 F=2: +6.76
$5P_{3/2}$	1.969(13)	0.870(22)	F=0: -6.30 F=1: -5.20 F=2: -2.13 F=3: +4.65
$6S_{1/2}$	21.89(9)	-	F=1: -27.36 F=2: +16.42
$6P_{1/2}$	4.05(7)	-	F=1: -5.06 F=2: +3.04
$6P_{3/2}$	0.866(8)	0.370(15)	F=0: -2.79 F=1: -2.29 F=2: -0.93 F=3: +2.04
$5D_{5/2}$	-0.24(7)	-	F=1: +1.26 F=2: +0.78 F=3: +0.06 F=4: -0.90
$5D_{3/2}$	0.44(10)	-	F=0: -1.65 F=1: -1.21 F=2: -0.33 F=3: +0.99
$6D_{5/2}$	-0.12(4)	-	F=1: +0.63 F=2: +0.39 F=3: +0.03 F=4: -0.45
$7D_{5/2}$	0.250(10)	0.050(20)	F=0: -0.88 F=1: -0.68 F=2: -0.23 F=3: +0.57

A, B: magnetic-dipole and electric-quadrupole constants (measured; see data file for sources).